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SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !



Diseases of Salivary Glands

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Outline:

- ▶ Introduction
- ▶ Anatomy and physiology of salivary glands
- ▶ Classification of salivary gland disorders
- ▶ Infections of salivary glands
- ▶ Common benign and malignant tumors
- ▶ Management and complications



Introduction

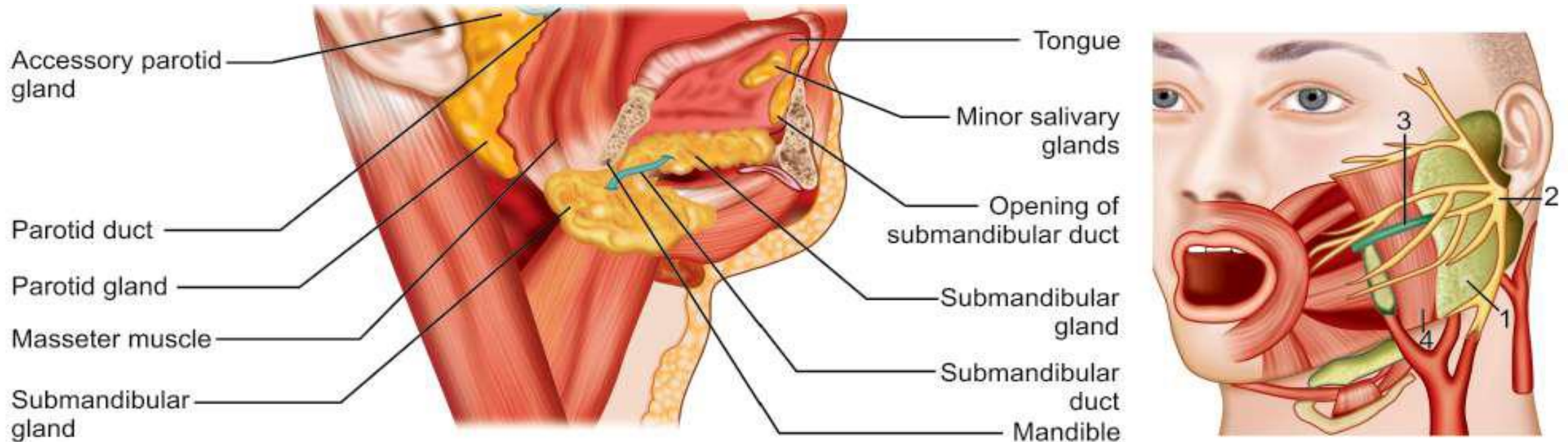
- ▶ Salivary glands are classified as **major & minor** glands
- ▶ Major salivary glands are paired glands:–
 - Parotid glands
 - Submandibular glands
 - Sublingual glands
- ▶ The numerous minor salivary glands are widely distributed in the oral cavity.
- ▶ Salivary glands secretion contains **water ,electrolytes ,urea ,ammonia ,glucose ,fats & proteins.**

Parotid gland

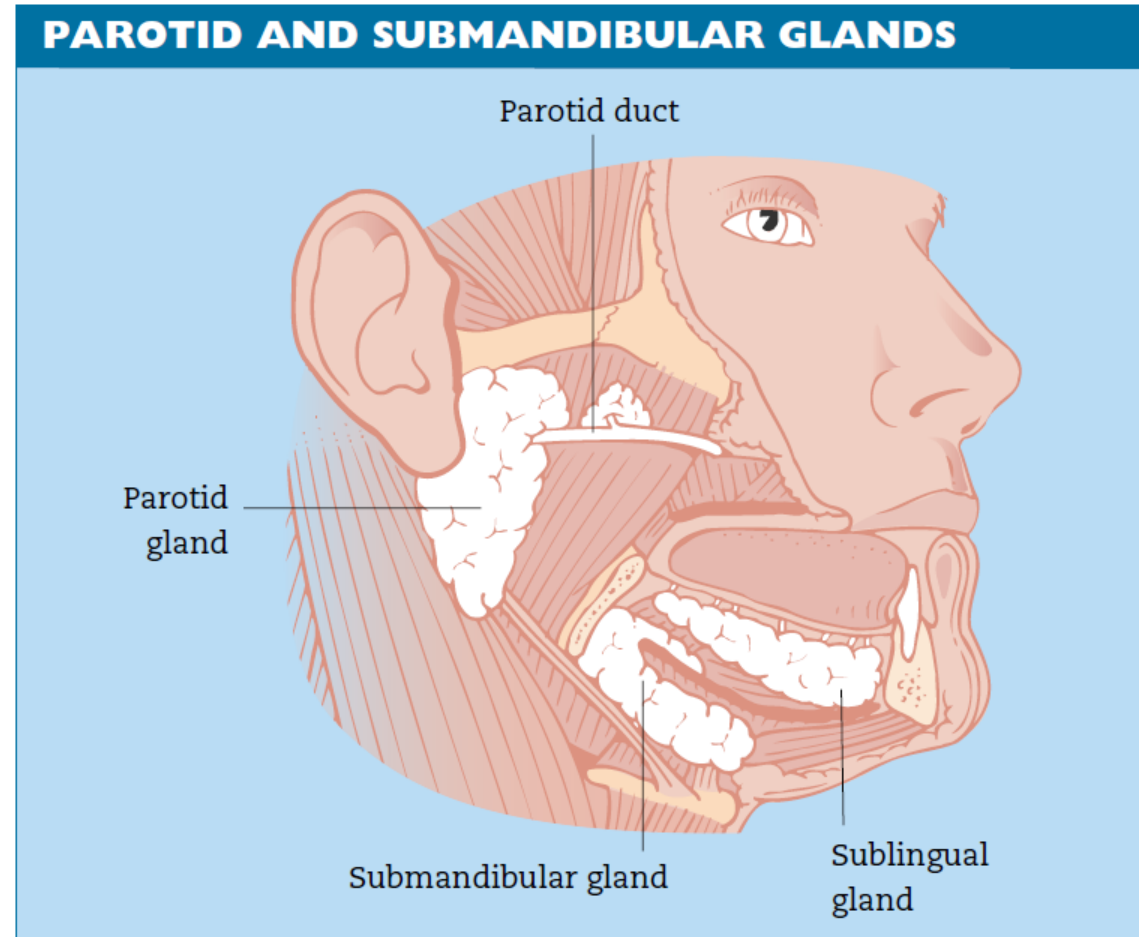
- ▶ It is the largest salivary gland.
- ▶ It is **Pyramidal** in shape.
- ▶ Two lobes **superficial & deep** connected by an isthmus at posterior part of the gland.
- ▶ Its **apex is toward angle** of mandible and **base at the external acoustic meatus**.
- ▶ Anteriorly gland extends up to buccal pad of fat.
- ▶ Posteriorly encircles posterior border of mandible.
- ▶ Parotid gland secretion is **serous** in nature.

Parotid duct (Stenson's duct)

- ▶ Stenson`s duct emerges at anterior part of gland.
- ▶ Stenson`s duct opening is seen as a papilla in the buccal mucosa opposite to **maxillary second molar**.



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Submandibular gland

- ▶ The gland is located in submandibular space.
- ▶ Extending inferiorly up to **digastric muscle**.
- ▶ Superiorly **mylohyoid muscle**.
- ▶ Posteriorly up to **angle of mandible**.
- ▶ Anteriorly **mid portion of body** of the mandible.
- ▶ Submandibular gland secretion is mixed.

Submandibular duct (Wharton's duct)

- ▶ The duct starts from deep part of the gland.
- ▶ Turns sharply at the posterior border of mylohyoid muscle anteriorly & superiorly, crosses hyoglossus muscle.

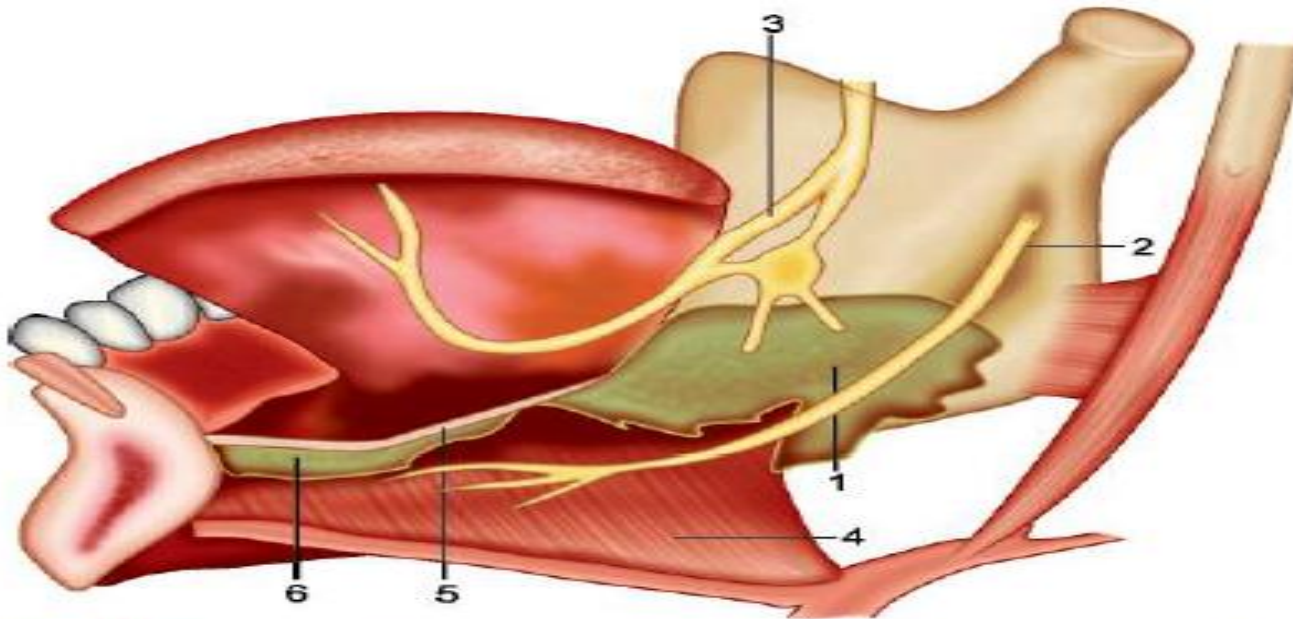


Fig. 37.2: Submandibular gland anatomy: (1) Submandibular salivary gland, (2) Hypoglossal nerve, (3) Lingual nerve, (4) Mylohyoid muscle, (5) Wharton's duct, (6) Sublingual gland

Sublingual gland

- ▶ This gland is located in sublingual space it is present in association with sublingual fold below tongue ,& divided into anterior & posterior part
- ▶ Sublingual gland secretes both **serous & mucous primarily mucous.**
- ❖ **Bartholin' s duct**
- ▶ The ducts of anterior part may join to form a large main duct called Bartholin' s duct.
- ▶ Revinus ?

Minor salivary glands

- ▶ More than **800** minor salivary glands may be present in oral cavity.
- ▶ Secrete **mucous** secretions.

Functions of saliva

- Digestive function
- Protective function
- Cleansing
- Lubrication
- Antibacterial action

Cont.

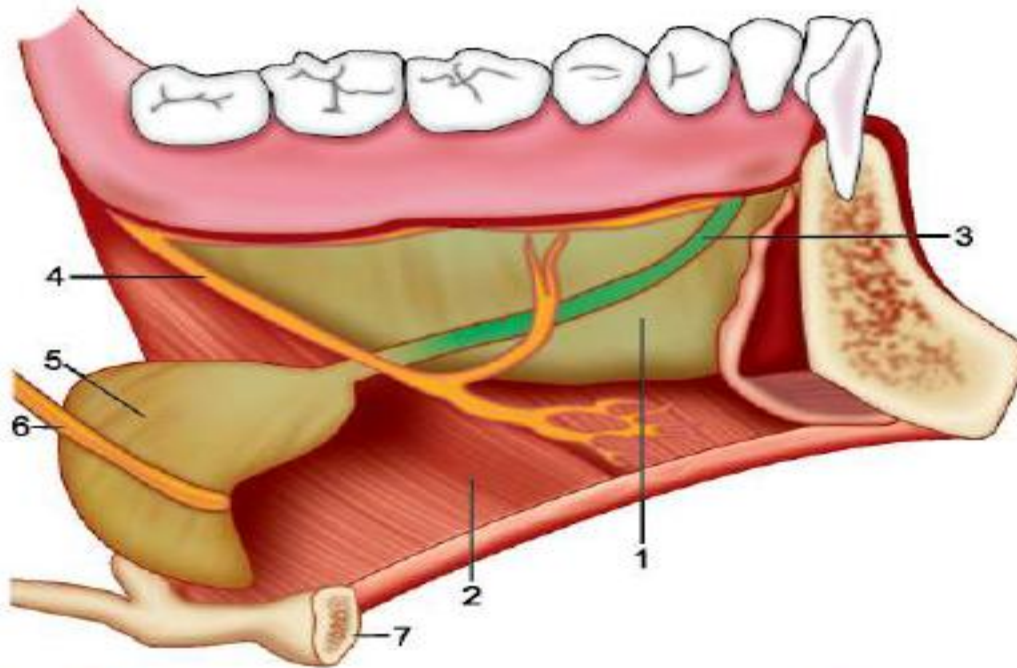


Fig. 37.3: Sublingual salivary gland anatomy: (1) Sublingual gland (2) Mylohyoid muscle, (3) Wharton's duct, (4) Lingual nerve, (5) Submandibular gland, (6) Hypoglossal nerve, (7) Hyoid bone

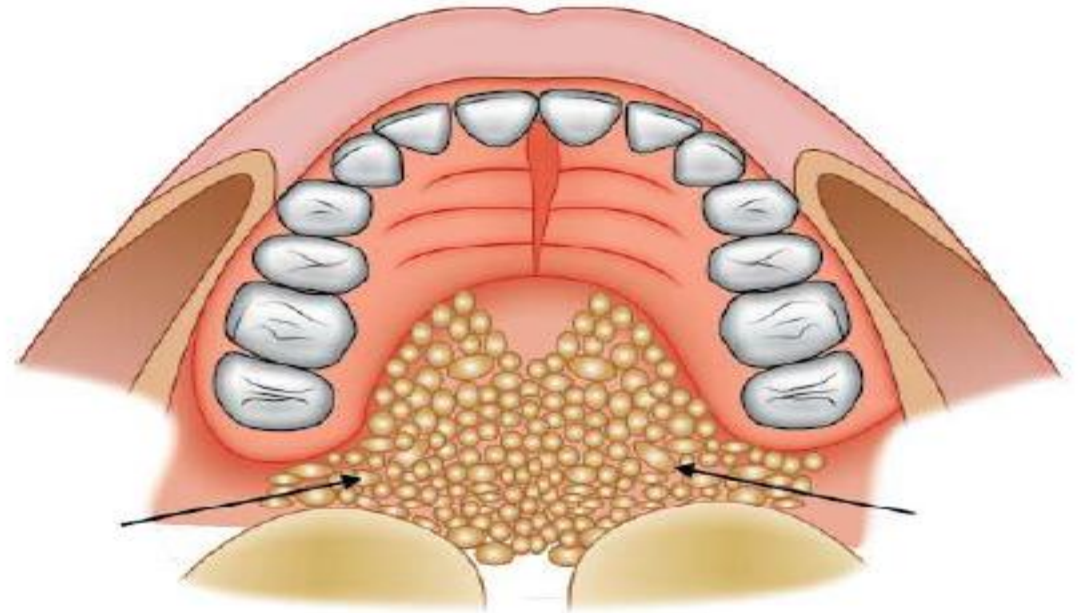


Fig. 37.4: Minor salivary glands in the palate

Classification of salivary gland diseases

- ▶ Salivary gland dysfunction

 - Xerostomia*

 - Sialorrhea*

- ▶ Developmental

 - Aplasia – absence of the gland*

 - Atresia – absence of the duct*

 - Aberrancy – ectopic gland*

Cont.

▶ Enlargement of the gland

Infectious

- Viral ; mumps
- Bacterial

Non – inflammatory

- Autoimmune; Sjogren's syndrome
- Alcoholic cirrhosis
- Diabetes mellitus

▶ Sialolithiasis

CONT.

▶ Cysts

Retention cysts

Extravasation cyst

Ranula

▶ Tumours of salivary glands

Benign tumours

- Pleomorphic adenoma
- Warthin's tumour

CONT.

Malignant tumours

- Mucoepidermoid carcinoma
- Acinic cell carcinoma
- Adenoid cystic carcinoma

▶ **Necrotizing sialometaplasia**

Xerostomia

- ▶ Xerostomia is a subjective sensation of a **dry mouth**.
- ▶ Affects women more than men and older people.

Causes

- ✓ Antihistamines, Decongestants
- ✓ Antidepressants ,Antipsychotics
- ✓ Antihypertensive, Anticholinergic
- ✓ Salivary gland aplasia
- ✓ Excessive smoking , mouth breathing
- ✓ Local radiation therapy , Sjogren's syndrome &HIV infection
- ✓ Aging

Cont.

Clinical features

Dry mouth with foamy ,thick ,& ropy saliva

Gloves stick to the mucosa

Difficulty in mastication & swallowing

More chance for candidiasis & caries

Treatment

Removal of the cause

Maintenance of oral hygiene

Use of sialagogues

Systemic pilocarpine 5-10mg--tid/qid

Topical fluoride application

Sialorrhoea

- Sialorrhoea is excessive salivation.

Causes

- ✓ local irritation like aphthous ulcers or ill-fitting dentures.
- ✓ Profuse salivation like in rabies.
- ✓ Heavy metal poisoning.
- ✓ Gastro esophageal reflux diseases.
- ✓ After certain medication like lithium & cholinergic agonists.
- ✓ Mental retardation in children.

Cont.

Treatment

- ▶ Removal of the cause.
 - ▶ Anticholinergic medication.
 - ▶ Submandibular gland resection.
 - ▶ Parotid duct ligation.
 - ▶ Ductoplasty

Sialadenitis

- ▶ Inflammation of salivary glands is known as **sialadenitis**.

Causes

- ▶ Viral infections
- ▶ Bacterial infections
- ▶ Allergic reactions
- ▶ Systemic diseases

Mumps

- ▶ It is also called as **epidemic parotitis**.
- ▶ It is caused by paramyxovirus and affects major salivary glands, especially the **parotid gland**.

Clinical Features:

The mumps virus can be transmitted through **urine, saliva or respiratory droplets**.

Incubation period is 16 to 18 days.

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Patients are contagious 1 day before & 14 days after the resolution.

Usually subclinical.

If symptomatic prodromal symptoms of **Low– grade fever, Headache, malaise & Myalgia.**

Discomfort & swelling over the lower $\frac{1}{2}$ of external ear down to posterior & inferior border of mandible.

Either one or both the parotid gland are enlarged and become tender.

Cont.

- ▶ Enlargement & pain are maximum in 2–3 days.
- ▶ Chewing mov't or saliva stimulating foods increases pain.
- ▶ Enlargement begins on one side & then extends to other.
- ▶ There may be edema & erythema involving the ductal orifice.

Cont.

Complications

Pancreatitis

Orchitis

Oophoritis

Meningio encephalitis

Diagnosis:

Urine, saliva & cerebrospinal fluid for culture.

Cont.

Treatment:

- Analgesics and antipyretics

- Bed rest

- Avoidance of sour foods

- Prior vaccination

Bacterial infection

- ▶ Bacterial sialadenitis affects parotid gland more.
- ▶ Submandibular glands are rarely affected.

Acute bacterial sialadenitis

- ▶ **Organisms**:- staph aureus , strep pyogenes, strep viridans.
- ▶ **Drugs** like tranquilizers; anti Parkinson drug ; diuretics; & antihistamines decrease salivary flow with increased chance of infection of salivary glands.

Clinical features

- ▶ Sudden onset of pain at angle of the jaw which is unilateral.

Cot.

- Affected gland is **enlarged & tender & extremely painful.**
- Inflammatory swelling is very tense & does not show much fluctuation.
- Skin is warm and red.
- Associated fever & trismus may be there.
- Purulent discharge from the affected duct orifice.

Histopathologic features

Accumulation of neutrophils is observed with in ductal system & acini.

Cont.



Fig. 37.10: Acute bacterial sialadenitis

Cont.

Treatment

- ▶ Antibiotics.
- ▶ Hydrating the patient.
- ▶ Stimulate the salivation by chewing sialagogues.
- ▶ Improve oral hygiene by debridement & irrigation.
- ▶ Surgical drainage if abscess is there.

Chronic Bacterial Sialadenitis

- It may be idiopathic or with factors like:–
 - ▣ Duct obstruction ,
 - ▣ Congenital stenosis,
 - ▣ Sjogren 's syndrome
- The microorganisms may be strep–viridans, e– coli

Clinical features

- Unilateral periodic pain swelling of the affected gland
- Gland may undergo atrophy which results in decreased salivary flow.

Cont.d

Histopathologic features

- ▶ Patchy infiltration of salivary parenchyma by lymphocytes & plasma cells.
- ▶ Atrophy of acini & ductal dilatation & sometimes fibrosis.

Sialography

- ductal dilatation proximal to area of obstruction.

Treatment

- ▶ Antibiotics
- ▶ Management of the underlying cause
- Excision of the gland.



Figure 11-20 • Chronic sialadenitis. Parotid sialogram demonstrating ductal dilatation proximal to an area of obstruction. (Courtesy of Dr. George Blozis.)

Sjogren syndrome

- ▶ Discussion ?
- ▶ Definition
- ▶ Types
- ▶ Diagnosis
 - Clinically
 - Investigation(laboratory, imaging &biopsy)
- ▶ Management

Sialadenosis

- ▶ It is **non-inflammatory, non-neoplastic swelling** of the salivary glands.
- ▶ Sialadenosis can occur in the following conditions:–
 - Hormonal disorders (pregnancy, hypothyroidism)
 - Diabetes mellitus
 - Alcoholic cirrhosis
 - Malnutrition
- ▶ Caused by dysregulation of **autonomic innervation** of salivary acini causing aberrant intracellular secretory cycle leading to excessive secretion of secretory granules causing enlargement of acini.

Clinical features

- ▶ Enlargement is usually painless
- ▶ Usually bilateral
- ▶ More common in women
- ▶ Commonly affects parotid glands

Histopathologic features

- ▶ Hypertrophy of acinar cells
- ▶ Nuclei are displaced to the base
- ▶ Cytoplasm is engorged with zymogen granules
- ▶ In DM & alcoholism – acinar atrophy & fatty infiltration

Treatment –Controlling underlying cause

Sialolithiasis

- ▶ Sialolithiasis is the formation of sialolith (**salivary calculi, salivary stone**) in the salivary duct or gland resulting in the obstruction of the salivary flow.

Sialolith

- ▶ Sialolith is a calcified mass with laminated layers of inorganic material from crystallization of salivary solutes
- ▶ The sialolith is **yellowish white** in color.
- ▶ Single or multiple, may be round & ovoid or elongated having size of 2cm or more diameter.

Cont.

- ▶ The minerals are various forms of calcium phosphate like **hydroxyapatite, octa-calcium phosphate** etc.
- ▶ Calcium & phosphorus ions are deposited on the organic nidus, may be desquamated epithelial cell, bacteria, foreign particle or product of bacterial decomposition
- ▶ It may be related to sialadenitis or ductal obstruction

Etiologies

- ▶ Discussion?

Clinical features

- Commonly seen in middle –age persons
- More common in submandibular salivary ductal system.
- Pain & swelling during & after eating food.
- Stone can be palpated if it is in the peripheral aspect of the duct.

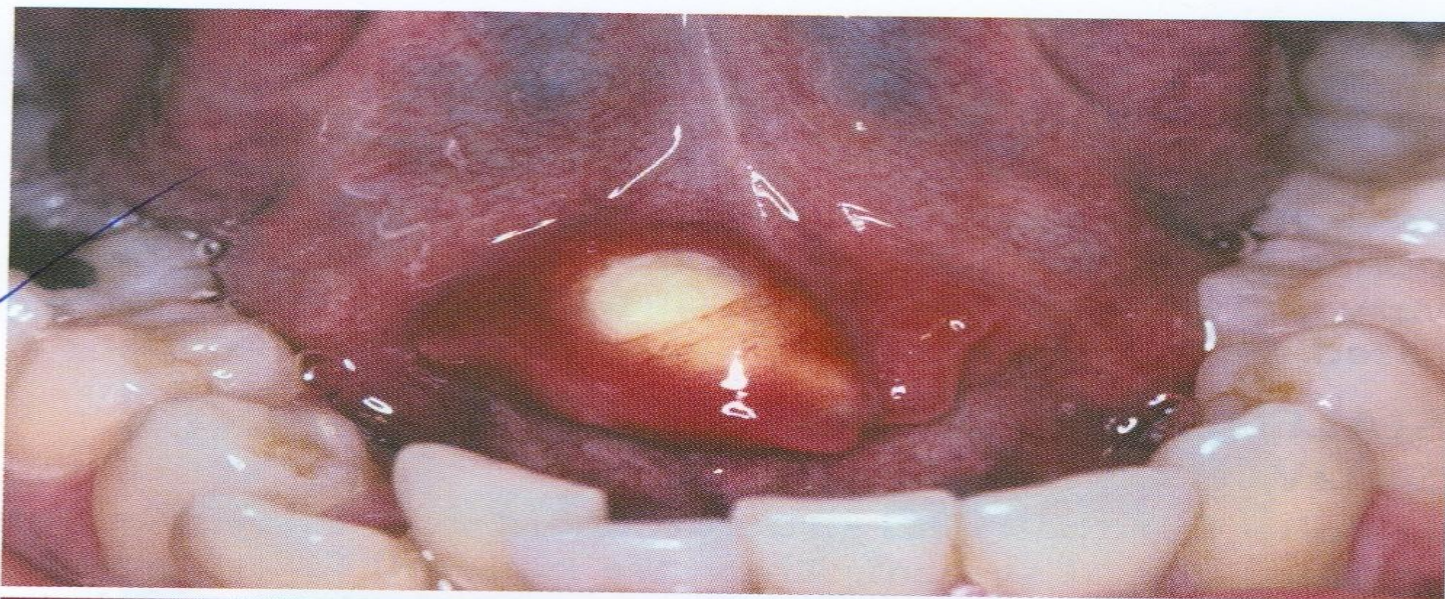
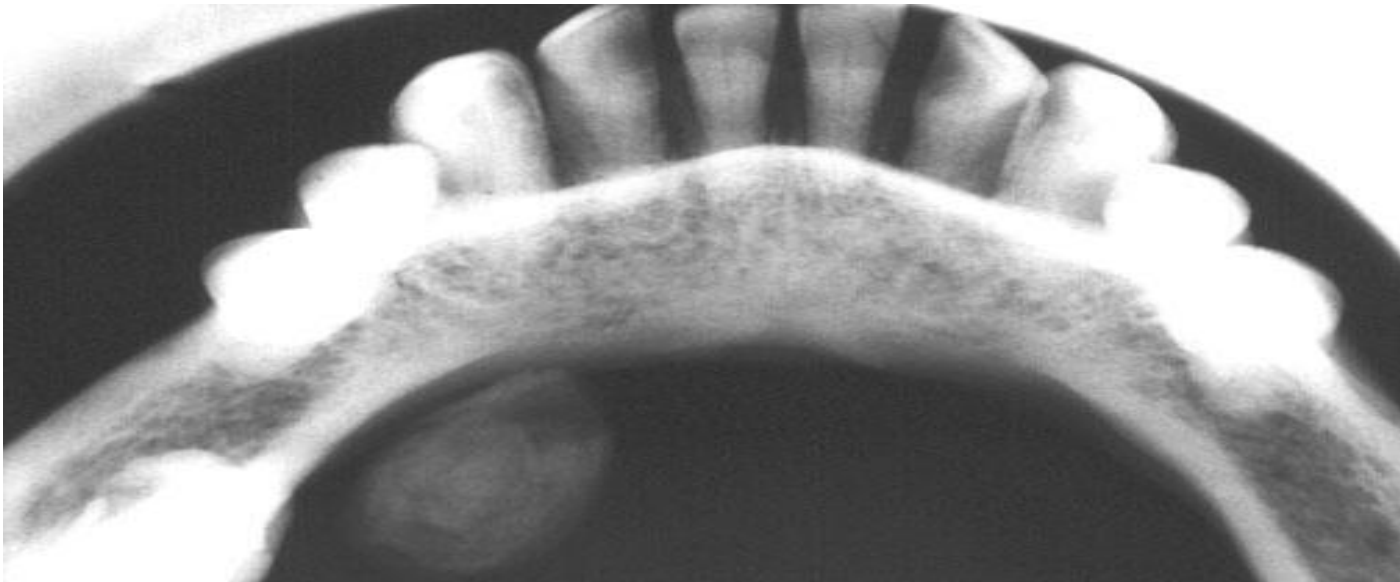


Figure 11 -12 ♦ Sialolithiasis. Hard mass at the orifice of Wharton's duct.

Investigations

- PA view , lateral oblique or occlusal view– shows radiopaque mass.
- U/S
- CT– Scan
- Sialography

- ▶ Panoramic x-ray–sialolithiasis of submandibular gland



Treatment

- ▶ Smaller sialolith which are located peripherally near ductal opening may be removed by manipulation called milking the gland.
- ▶ Piezoelectric shock wave lithotripsy.
- ▶ Surgically

Mucocele

- ▶ Lower lip is commonly affected.
- ▶ Other common sites are buccal mucosa, ventral tongue, floor of mouth.
- ▶ It can be superficial or deep.
- ▶ Superficial – elevated well circumscribed bluish vesicle
- ▶ Deep – nodule with no change in color.
- ▶ Cystic contents – thick mucous material.
- ▶ Usually covered by mucous membrane.
- ▶ There may have periodic rupture of the swelling releasing the contents.



Figure 11-1 • Mucocele. Blue-pigmented nodule on the lower lip.



Figure 11-2 • Mucocele. Nodule on the posterior buccal mucosa.

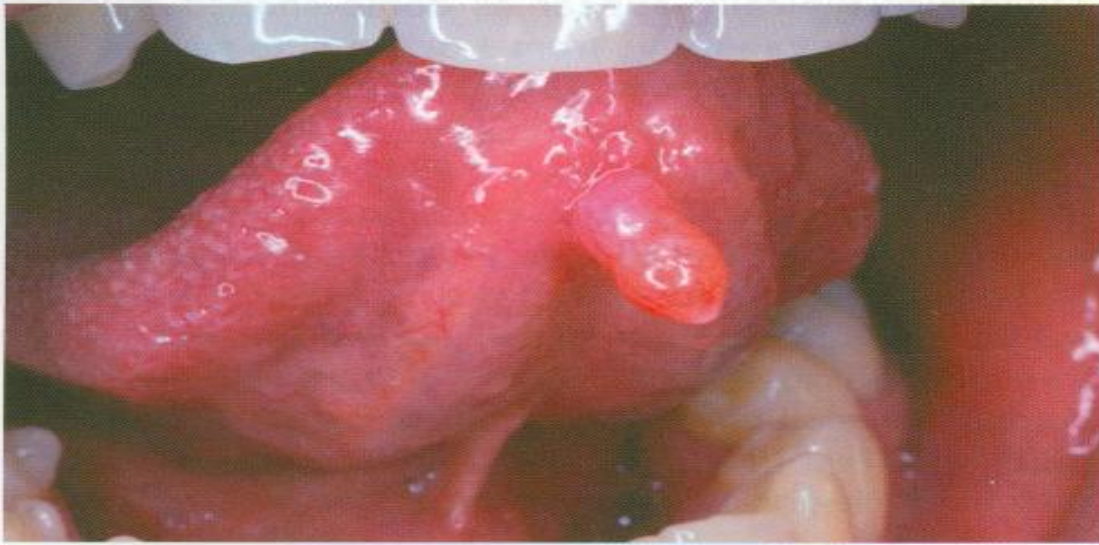


Figure 11-3 • Mucoccele. Exophytic lesion on the anterior ventral tongue.



Figure 11-4 • Superficial mucoccele. Vesicle-like lesion on the soft palate.

Cont.

- ▶ After rupture it may leave shallow painful ulcers.
- ▶ Some lesions resolve by itself.
- ▶ Histopathologic features
 - Area of spilled mucin surrounded by granulation tissue.
 - Adjacent minor salivary glands contain c/c inflammatory infiltrate.
- ▶ Treated by excision along with adjacent minor salivary glands to prevent recurrence.

Salivary duct cyst

- ▶ Mucus retention cyst or sialocyst which is **true cyst**.
- ▶ Epithelium lined cavity that arises from **salivary gland tissues**.
- ▶ May be caused by **ductal dilatation or secondary to ductal obstruction**.
- ▶ It can be seen in major or minor salivary glands.
- ▶ Cysts of major glands are common in parotid gland.
- ▶ Intraoral cyst are common in buccal mucosa, floor of mouth & lips.

Cont.

- ▶ They are **soft, fluctuant, asymptomatic swelling** & may appear bluish depending on the depth.

Histopathologically:–

- ❑ cyst may be lined by cuboidal, columnar or squamous epithelium surrounding the mucoid secretion in lumen.

Treatment

- ▶ Treated by local excision for minor salivary gland ducts.
- ▶ For major salivary glands total or partial removal of gland can be done.
- ▶ Sialagogues can stimulate salivation & prevent accumulation of mucus.

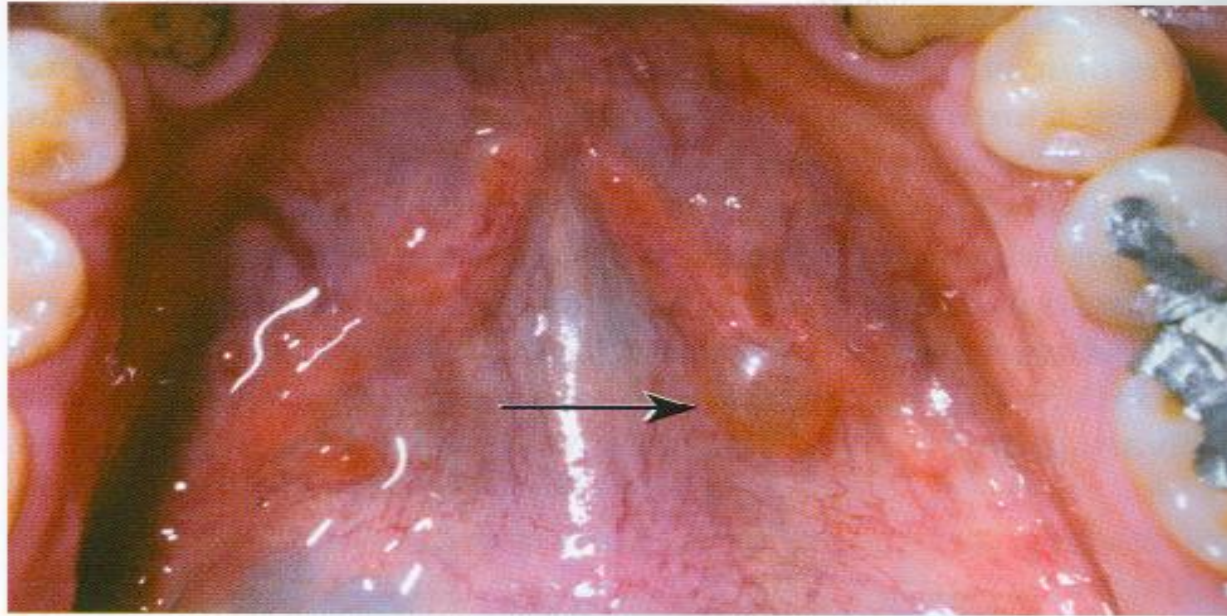


Figure 11-9 • Salivary duct cyst. Nodular swelling (*arrow*) overlying Wharton's duct.

Ranula

- ▶ Extravasation cyst usually arises from ducts of sublingual glands.
- ▶ Bluish, dome shaped, fluctuant swelling in floor of mouth.
- ▶ May enlarge and raise the tongue.
- ▶ Usually seen lateral to midline.
- ▶ May extend to the neck behind the posterior border of mylohyoid muscle (plunging ranula).

Treatment

- ▶ Excision of the sublingual gland.



Figure 11-7 • Ranula. Blue-pigmented swelling in the left floor of the mouth.

Pleomorphic adenoma

- ▶ It can affect both major & minor salivary glands.
- ▶ It commonly affects the parotid gland.

Clinical features

- ▶ More common in females.
- ▶ Small painless nodule at the angle of mandible or beneath the ear lobe.
- ▶ Well circumscribed ,encapsulated ,firm in consistency & may show area of cystic degeneration.
- ▶ Difficulties in mastication & talking.
- ▶ Initially tumor is movable but later becomes less mobile.

Cont.

- ▶ If deep lobe is affected ,a swelling in the lateral pharyngeal wall or soft palate.
- ▶ Minor salivary glands involvement is common in palate & lips as smooth surfaced dome shaped swelling.

Histopathologic features

- ▶ Well-circumscribed ,encapsulated tumor.
- ▶ Tumor is composed of a mixture of glandular epithelium & myoepithelial cells with in a mesenchyme like background may be myxoid or chondromatous or hyalinized.

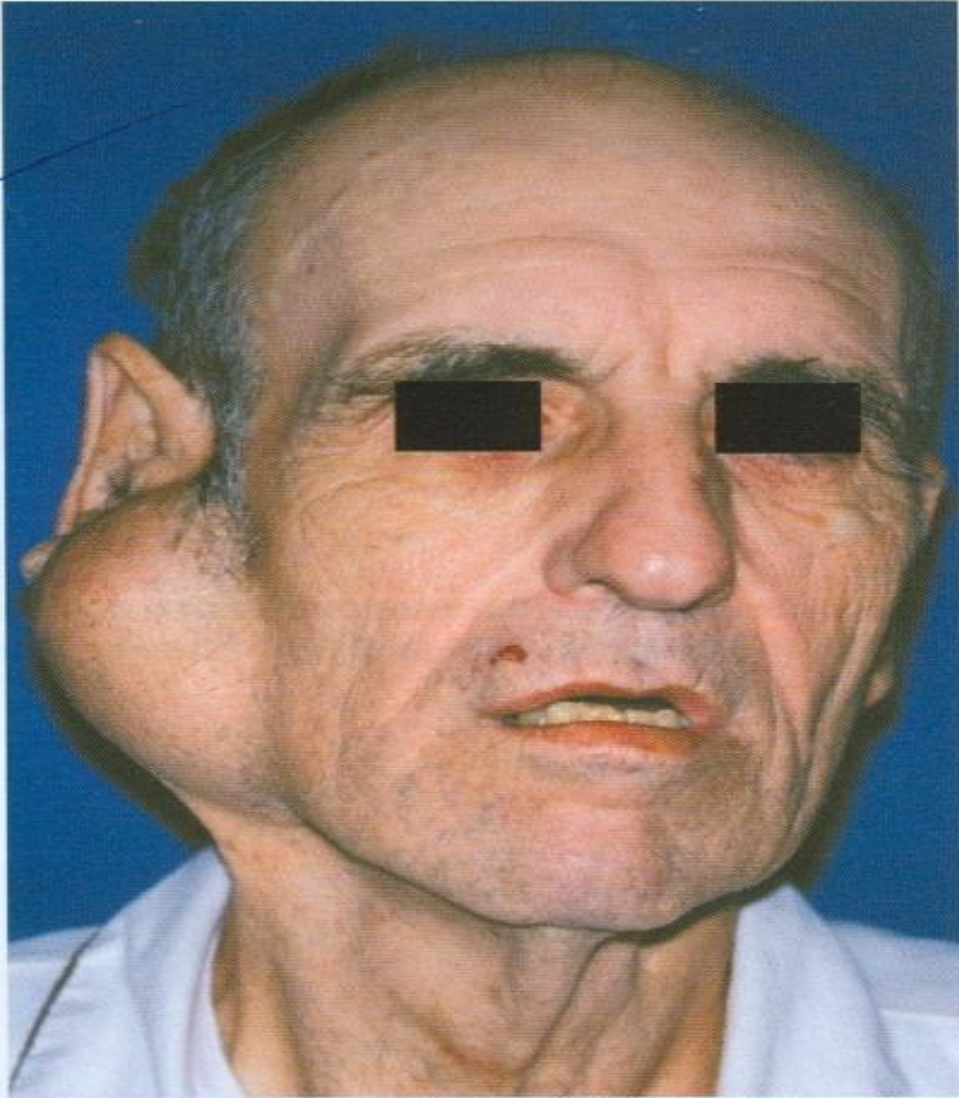


Figure 11-33 ♦ Pleomorphic adenoma. Slowly growing tumor



Figure 11-36 ♦ Pleomorphic adenoma. Firm mass of the hard palate lateral to the midline.

Cont.

- ▶ **Treatment** – surgical excision(parotidectomy)?
–complications?

Warthin tumor

- ▶ Papillary cyst adenoma lymphamatosum.
- ▶ Affects the parotid glands.
- ▶ Males are affected more.
- ▶ More common in smokers

Clinical features

- ▶ Firm or fluctuant, non– tender ,circumscribed mass in the region of angle or ramus of the mandible or beneath ear lobe.
- ▶ Common in the tail of the gland.
- ▶ Both sides of parotid gland affected.



Dr. Demerew Dejene (Oral & Maxillofacial Surgeon)

Cont.d

Histopathologic features

- ▶ Tumor composed of mixture of ductal epithelium & lymphoid tissue.

Treatment

- ▶ Surgical excision

Mucoepidermoid carcinoma

- ▶ Discussion?
- ▶ The low grade tumor behaves almost like a benign tumor with very good prognosis.
- ▶ High grade tumor behaves very aggressively.
- ▶ It occurs with equal distribution between males& females

Clinical features

- ▶ More common in parotid gland.
- ▶ It may grow slowly or rapidly.
- ▶ Painless swelling.
- ▶ Ulceration.

Cont.

- ▶ Facial paralysis
- ▶ Minor salivary gland tumors are common in palate & may have bluish hue.
- ▶ Local destruction & metastasis to regional lymph nodes & distant metastasis to the lungs.
- ▶ **Histopathologic features**
- ▶ Mucus producing cells & squamous cells.
- ▶ High grade tumors have cellular atypia.

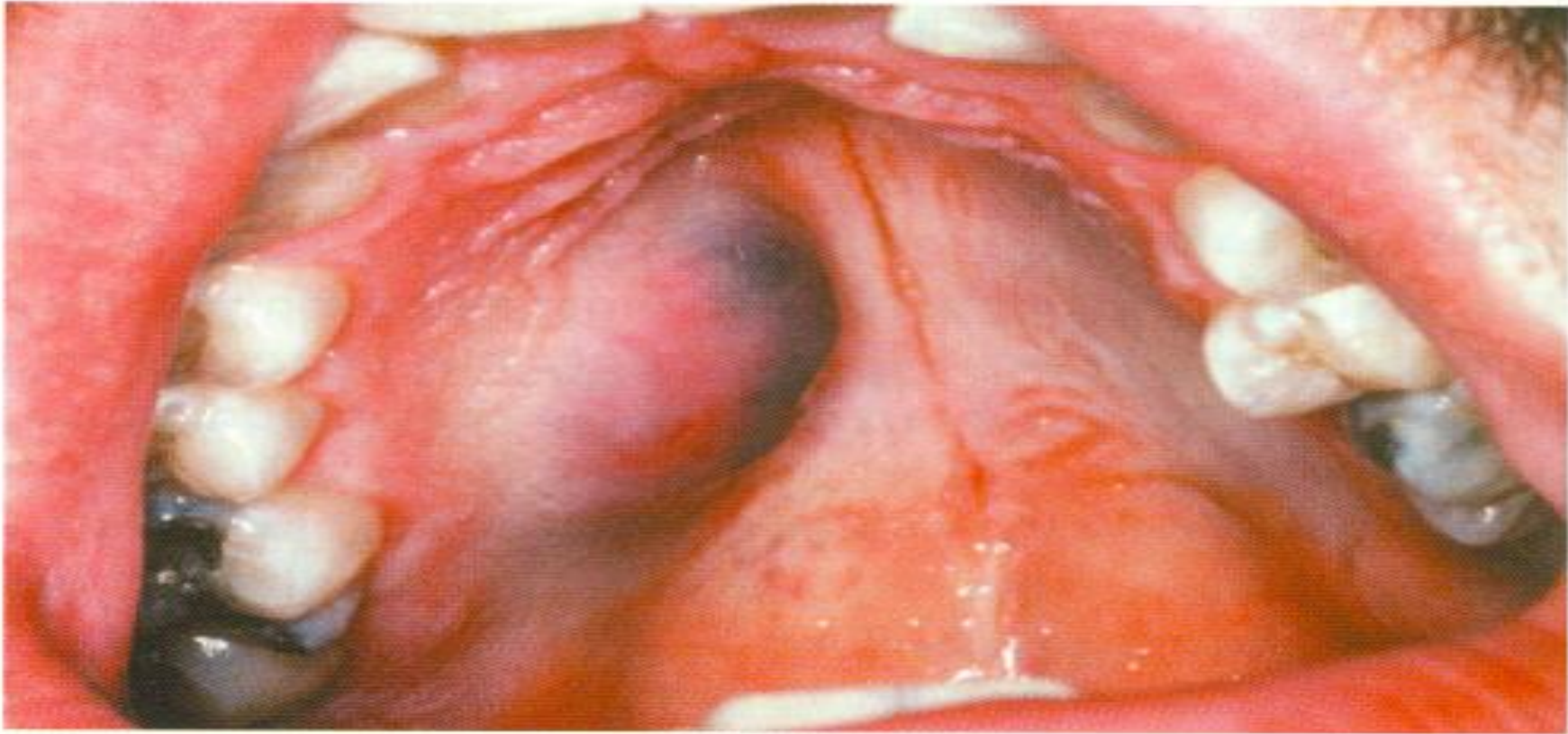


Figure 11-54 • Mucocystic carcinoma Blue-pigmented

Cont.

Treatment

- ▶ Surgical excision
- ▶ For minor salivary glands excision with surrounding normal tissues.
- ▶ For tumors with metastasis radical resection with radiation.

Acinic cell carcinoma

- ▶ A low grade malignancy.

Clinical features

- ▶ Commonly occurs in parotid gland
- ▶ Common in females
- ▶ Usually asymptomatic
- ▶ Commonly affects serous acini
- ▶ In minor salivary glands it is common in buccal mucosa, lip & palate.
- ▶ It may be a slow growing swelling
- ▶ Sometimes pain, tenderness may be there



Cont.

Histopathologic features

- ▶ Acinar cell has abundant granular basophilic cytoplasm & round, darkly stained eccentric nucleus

Treatment

- ▶ Tumor confined to the superficial lobe is treated by lobectomy
- ▶ Tumor involving deep lobe – parotidectomy
- ▶ Radiotherapy for severe cases

Adenoid cystic carcinoma

- ▶ It is also called cylindroma

Clinical features

- ▶ Slow growing swelling
- ▶ Commonly occurs in palatal minor salivary glands
- ▶ Commonly occurs in middle aged individuals
- ▶ Constant ,low grade, dull aching pain
- ▶ Facial nerve paralysis in parotid tumors

Histopathologic features

- ▶ Islands of basaloid epithelial cells that contain multiple cylindric ,cyst like spaces.



Figure 11-60 Adenoid cystic carcinoma. Painless mass of the

Cont.

- ▶ Perinuerual invasion

Treatment

- ▶ Surgical excision

Necrotizing sialometaplasia

- ▶ It is a locally destructive inflammatory lesion affecting **minor salivary glands**.
- ▶ Causes?
 - Ischemia of salivary tissues

Clinical features

- ▶ Commonly occurs in men
- ▶ Mostly affects minor salivary glands of the palate but can affect lip or retro molar pad minor salivary gland
- ▶ The lesion occurs as a swelling with paresthesia then it sloughs leaving large ulcer or ulcerated nodule
- ▶ Edge of lesion presents with an inflammatory reaction



Figure 11 -31 ♦ Necrotizing sialometaplasia. Later-stage lesion

Cont.

Histopathologic features

- ▶ Acinar necrosis
- ▶ Squamous metaplasia of salivary ducts

Ddx–SCC

Treatment

- ▶ Debridement & irrigation
- ▶ The lesion is self – limiting one & heals in 6 to 8 weeks

Complications of surgery of salivary glands

- ▶ Damage to lingual nerve
- ▶ Hypoglossal nerve
- ▶ Damage to Wharton's duct
- ▶ Damage to Auriculotemporal nerve
- ▶ Facial nerve paralysis

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The End

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Thank You